

SA Monitor for Remote Operators

How each component is calculated, and what it is for

A passive, behaviour-only situation-awareness monitor for the remote operation of a live vehicle. Built and validated on 439 probes from 37 operators.

 *green · amber · red the operative state read from behaviour*

A two-layer design

The monitor separates two questions that a single per-probe alarm would blur together.

● Diagnosis

what is happening now

- Two objective risk dials, error and latency, read continuously from behaviour.
- A manual confidence check the operator answers only when prompted.
- A probe timeline and a plain-language diagnosis for the selected operator.

● Triage

who needs attention

- A per-operator ranking that orders the whole team by calibrated risk.
- A drill-in into any operator's probe history and events.
- The same evidence as the diagnosis layer, aggregated per operator.

Deployment framing. The study is probe-resolution: each estimate is formed in the 5 s window before a query. A deployed monitor would carry no SA probes. It would stream the same gaze and telemetry features continuously and hold a rolling estimate, for which the trailing average used here stands in.

The two objective dials

ERROR

The modelled probability that the operator's answer to the current query is wrong.

LATENCY

The modelled probability that the response is delayed, taken as 3.5 s or longer.

Inputs

Eleven pre-probe features from the 5 s window before each query.

Gaze: target-object dwell, road-gaze proportion, saccade rate and velocity, scanpath rate.

Control: steering and speed variance, steering-reversal rates, throttle-release.

A third 'unsure' dial was retired. It was the weakest detector and largely redundant with error. Confidence resists prediction from behaviour, so it is captured directly from the operator instead.

Each dial is read against its cohort base rate: error 0.22, latency 0.14 (the white reference line on the gauge).

How a dial value is calculated

1

Features

Eleven pre-probe gaze and control measures for the current query.

2

Grouped out-of-fold model

Error by gradient boosting, latency by random forest. Every probe is scored by a model trained on other operators (GroupKFold).

3

Isotonic calibration

The model's ranking is mapped to a true probability, fitted out of fold so it never sees the probe it scores.

4

Cap at 0.90

No gauge reads as a certain failure: the top calibration step rests on very few cases.

Result: a calibrated probability on the same 0 to 1 scale as the failure base rate.

Out-of-fold discrimination (ROC-AUC)



A soft aid, not a verdict. Discrimination is modest, so the dials bring attention forward rather than judging a single probe. ROC is reported on the raw out-of-fold scores, since in-sample calibration would inflate it.

From single probe to recent state

Each dial is smoothed along the operator's probe sequence. The gauge shows this recent state, not the single latest probe and not a whole-session mean.

```
recent_t = 0.5 * dial_t + 0.5 * recent_(t-1)
```

```
exponential moving average: each probe back counts  
half the one after it (half-life 1 probe)  
span 3 -> alpha = 2 / (span + 1) = 0.5
```

 **single probe:** too noisy

 **session mean:** too lagging

 **recent state:** responsive, yet stable

Trend arrow is the change from the previous probe's recent value (rising risk, or settling). **Persistence badge** marks a caution that has held for two probes in a row, so a diagnosis reflects a sustained state rather than one reading.

Smoothing deliberately compresses the scale, which is why the traffic-light thresholds are set on the smoothed band and not on the raw per-probe value.

Setting the Amber and Red bands

The traffic light reads the smoothed composite, so the cut-points are set on that band's own distribution. A precision-recall curve of any-failure against the band fixes two operating points.

0.19

Amber an early, sensitive caution

Chosen to recall about 85% of failures. It flags most attention dips, and accepts more false alarms in return.

0.28

Red a later, higher-precision warning

Chosen to recall about 50% of failures. Fewer probes reach it, and those that do are more genuinely risky.

Why on the band, not the probe. The averaging compresses the scale. Cut-points set on the per-probe composite and applied to the smoothed band would not target the same recall, so both are fixed on the band the traffic light actually reads.

The triggered confidence check

A manual confidence rating, 1 to 7, that the operator gives only when prompted, about three times each (roughly a quarter of probes). Prompts fire at scheduled points (an early baseline and a mid-run check) and on any Red warning that persists for two probes.

Confirmed

Rating 4 or below with objective risk elevated. The operator agrees something is off. About 66% of confirmed-Red probes are true failures.

Overconfident

Rating 6 or above with objective risk elevated. The warning is not acknowledged: it stands, and a flag is raised.

Scheduled-low

Rating 4 or below while the dials are quiet. An aware risk the objective dials missed. About 57% fail.

Injection. A confirmed or scheduled-low reading raises that probe to the confirmed-failure level (about 0.66) before smoothing. The recent-state average then carries it forward and lets it decay, so an acknowledged deficit persists rather than vanishing at the next probe.

Two hard rules

● Overt lapse forces Red

A hazard encountered before the query is a categorical event, not a calibrated probability.

```
lapse_inj = 0.90
```

Injected into the series, then carried and decayed by the average like any other reading.

● Overconfidence hold

An unacknowledged objective-Red does not decay below Red while the operator stays confident.

a floor on the displayed state, not an injection

It holds the current reading without inflating later probes, since it never enters the average. Overconfident probes are about twice as enriched for real hazards.

Both rules express the same principle: a categorical hazard or an unacknowledged risk should outrank a quiet soft dial.

The operative recent risk

```
attn_risk      = max( error_dial , latency_dial )  
risk_aug       = max( attn_risk , confidence_inj , lapse_inj )  
recent_risk    = EWMA( risk_aug , span 3 )
```

The worst of the two dials sets the objective risk. The confidence check and the lapse rule can raise a probe above that. The smoothed result is the **operative recent risk**: the single state the traffic light, the timeline band, and the diagnosis text all read.

Objective view. The two dials shown on the gauges use the same average without the injections. The composite band adds the confidence and lapse escalations on top, which is why the band can sit above either single dial.

Operator 18_RT, at one moment

At the current probe the objective dials are quiet, but a confirmed low-confidence check has just fired. Three numbers coexist, and all three are correct.

0.31

Single probe

the worst dial at this probe alone

0.25

Recent state, objective

the smoothed dials, no confidence check

0.50

Operative recent risk

once the confirmed check is included

How 0.50 arises. The confirmed check injects 0.66 at this probe. Averaged with the recent history (0.5 of 0.66 plus 0.5 of the previous 0.34) this gives 0.50, which sits above the Red line at 0.28. The objective dials, without the check, read 0.25. The single worst dial at this probe alone is 0.31.

Ranking the team

The board orders every operator by a single calibrated triage score. The bar colour shows whether error or latency dominates, and any operator can be opened for drill-in.

How the score is built

- Each probe is scored as the worst of the objective dials and any confidence or lapse event.
- Each event is entered at its own empirical failure rate, not the 0.90 display level.
- The operator's score is the average across their probes.

Why calibrated weights

Entering events at their empirical rates, rather than the display level, stops rare hazards dominating the mean. It raises the correlation between operator rank and actual failure rate.

Spearman 0.58 → 0.66

One ranking, from the same evidence as the live view

Objective dials, overt lapses, and confidence checks are combined once, at calibrated weights, so the supervisor's ordering matches how often operators actually fail.

What the monitor is, and is not



Modest discrimination. The dials sit at ROC about 0.65 to 0.70. They bring attention forward; they do not judge any single probe.



Small-sample calibration. Calibration rests on few cases, so the top of the scale is capped and no gauge reads as a certain failure.



Sample-specific thresholds. Amber and Red are recall-targeted on this sample and would be re-fitted for a deployment.



Cohort, not personal, baseline. The reference line is a group base rate. Per-operator baselines are a deployment improvement, not yet fitted.



Probe-resolution validation. The study uses SA probes as a scaffold. A deployment would stream the same features continuously, with none.

THE MONITOR IN ONE LINE

Behaviour becomes calibrated dials; the dials become a recent state; the recent state becomes a traffic light; confidence checks and hard rules escalate it; and the team is ranked by the same evidence.



Towards Passive Measurement of SA in the Remote Operation of Live Vehicles

Chapter 5: Jefferies, R., Durant, S., Guy, I., Helman, S., & Dalton, P. (2026). Keeping an eye on remote operators: A passive, real-time situation awareness indicator from gaze and vehicle control during live remote driving (In preparation).